

SMART INDIA HACKATHON 2026



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HACKATHON
2026

TITLE PAGE

- **Problem Statement ID –26036**
- **Problem Statement Title- Development of an Online Verification System for Weighing and Measuring Instruments**
- **Theme-Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name-HexaCore**





- Unified digital platform for the complete lifecycle of weighing and measuring instruments, from registration and verification or re-verification to certificate issuance.
- Businesses can register instruments and track applications online.
- Legal Metrology Officers can conduct structured inspections through an offline-first field app, capturing measurements, photographs, GPS and time evidence, even in low-connectivity areas.
- Verified instruments receive cryptographically protected digital certificates with QR codes for instant public verification of authenticity and validity.
- Automated expiry alerts and a Digital Instrument Passport preserve each instrument's complete verification history.
- Combines offline field operations, evidence traceability, tamper-evident certificates, QR-based public trust and persistent digital lifecycle management in one platform.



Digital Instrument Passport

Complete lifecycle/
history of every
instrument.



Offline Field Verification

Inspect in low-
connectivity areas and
sync later.



Tamper-Evident Certificates

SHA-256 integrity +
RSA-PSSS digital
signature.



Public QR Verification

Instantly verify
certificate status
without login.



AI-assisted Inspection

OCR/evidence-
quality/ anomal
assistance



Automated Alerts

Expiry alerts &
reminders to ensure
compliance.

Key Innovation

- ✓ **Digital Instrument Passport:** complete lifecycle/history of every instrument.
- ✓ **Offline Field Verification:** officers can inspect in low connectivity areas and sync later.
- ✓ **Tamper-Evident Certificates:** SHA-256 integrity +RSA-PSSS digital signature.



1 Registration

Business registers
instrument & submits
application.



2 Application & Assignment

Application reviewed
and assigned to an
officer.



3 Scheduling

Inspection scheduled
as per availability and
location.



4 Field Inspection

Officer inspects evidence
capture, photos, GPS,
time, measurements.



5 Review & Decision

Officer reviews
evidence and takes
final decision.



6 Digital Certificate Issued

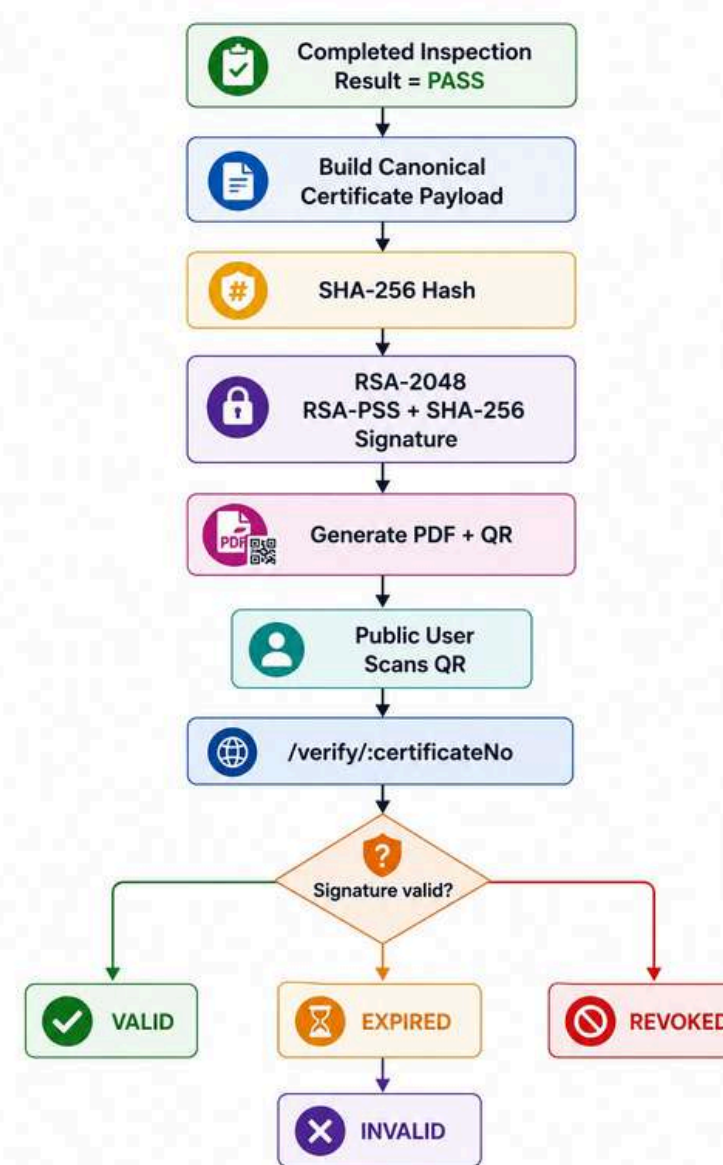
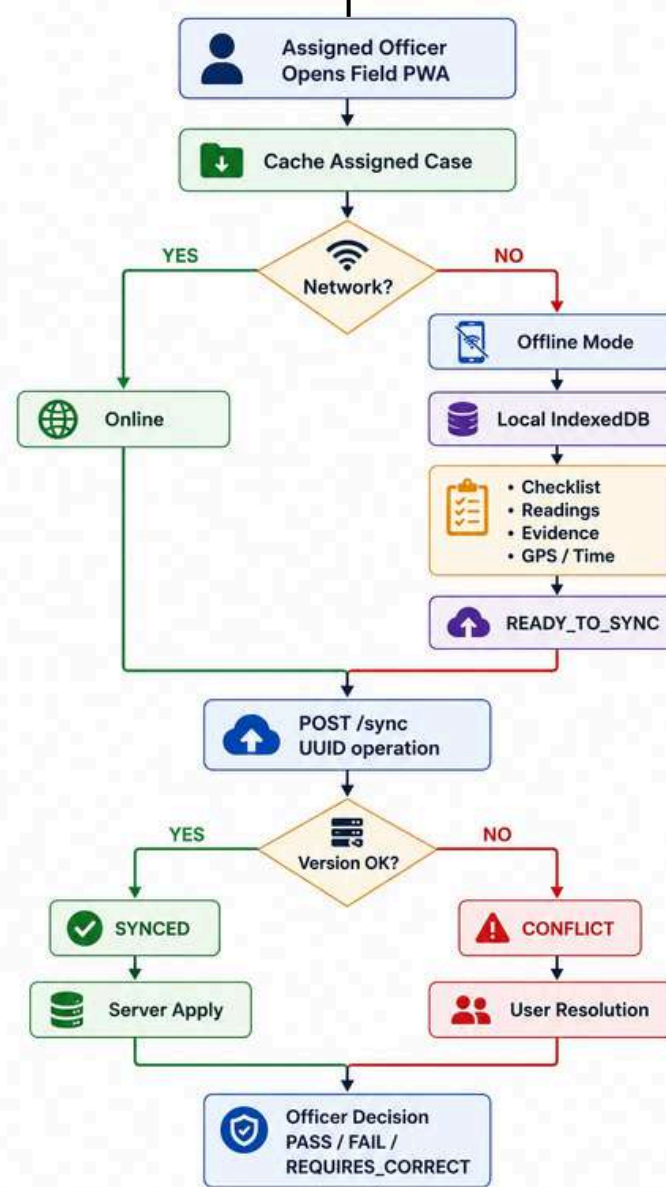
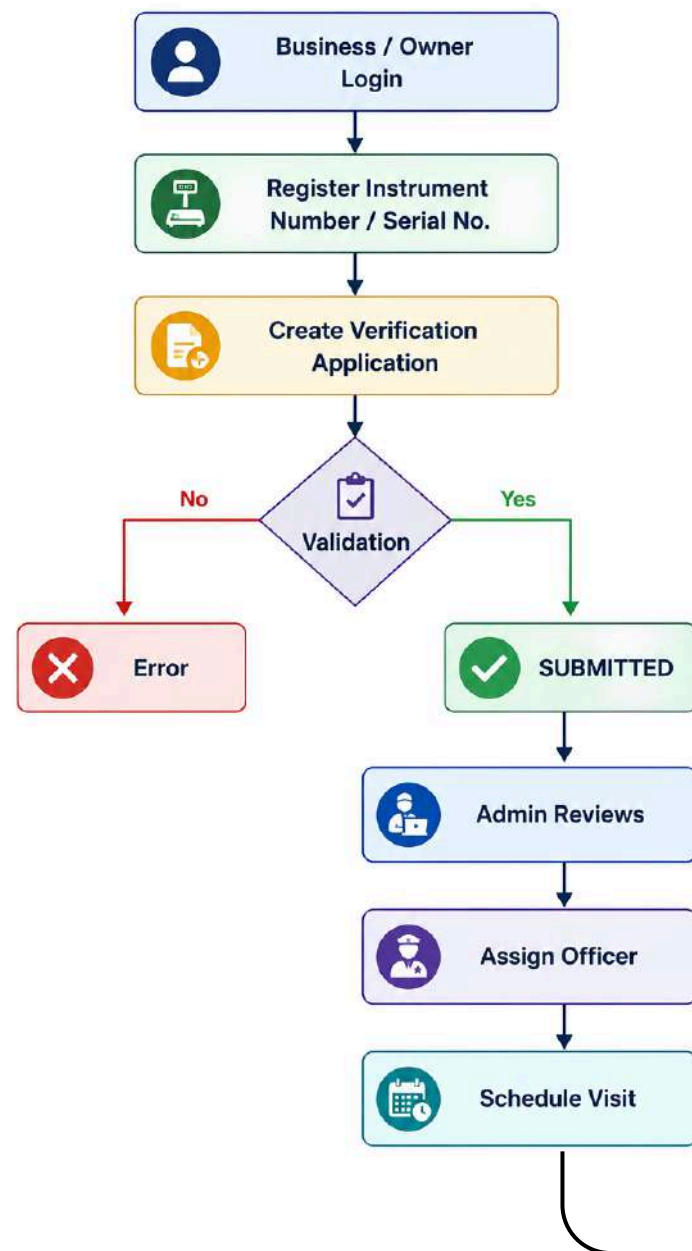
Tamper-evident
certificate with QR
code is generated.



7 Public Verification

Anyone can scan QR
to verify authenticity
and validity.

Flowchart



TECH STACK

FRONTEND (WEB)


Next.js
TypeScript
Tailwind CSS
shadcn/ui

MOBILE (OFFICER FIELD APP)


Flutter
Dart
Hive
Connectivity Plus

BACKEND (API & SERVICES)


Django
Python
JWT Authentication
Django REST Framework

DATA & INFRASTRUCTURE


PostgreSQL
Redis (Cache / Sessions)
MinIO (S3 Compatible)
Cloudinary (Media Storage)

FEASIBILITY AND VIABILITY

ECONOMIC

- Cost-effective:**
Open-source technologies reduce licensing and development costs.
- Digital-first:**
Reduces paperwork, manual processing and administrative overhead.
- Automation:**
AI-assisted OCR and evidence checks reduce repetitive workload.
- Low infrastructure cost:**
docker-based deployment can run on a single environment initially.
- Scalable model:**
Cloud-ready architecture enables expansion with low incremental cost.

TECHNICAL

- Proven stack:**
React, TypeScript, NextJs, PostgreSQL, MinIO, Docker.
- Prototype-ready:**
Core workflows can be demonstrated using synthetic data.
- Offline-first:**
PWA + IndexedDB enables field verification in low-connectivity areas.
- Secure by design:**
SHA-256, RSA-PSS, JWT/RBAC and QR-based verification.
- API-ready:**
REST APIs enable future integration with government/enterprise systems.

SCALABILITY

- Modular architecture:**
Features can be extended without redesigning the complete system.
- Configurable rules:**
Verification workflows can adapt to different jurisdictions.
- Cloud-ready:**
Containerized deployment supports migration from single-server to cloud infrastructure.
- Digital Instrument Passport:**
Persistent lifecycle records support large-scale instrument management.
- Future-ready:**
Architecture can evolve toward higher loads, integrations and analytics.

CHALLENGES

- Different state/legal workflows
- Poor field connectivity
- Certificate tampering
- Sensitive data/security
- AI false positives
- Government integration availability
- Incorrect legal assumptions

SOLUTIONS / STRATEGY

- Configurable, versioned workflow & rule engine
- Offline-first PWA + encrypted local queue + idempotent sync
- SHA-256 hash + RSA-PSS signature + server-side verification
- RBAC, JWT, Argon2id, encryption, audit logging
- AI remains advisory, authorized officer makes final decision
- API adapter architecture + synthetic/mock data for prototype
- Statutory tolerances/validity remain configurable until validated

- Reliable**
- Cost Effective**
- Scalable**
- Secure**

“ One instrument, One digital identity, One trusted lifecycle ”



IMPACT ON STAKEHOLDERS



FOR BUSINESSES

- Faster online application & status tracking
- Digital access to certificates
- Automated expiry reminders
- Reduced paperwork and physical follow-ups



FOR LEGAL METROLOGY OFFICERS

- Structured digital inspections
- Mobile/offline field operations
- Faster evidence capture & reporting
- Complete instrument history at the point of inspection



FOR GOVERNMENT

- Centralized instrument registry
- Real-time tendency & expiry monitoring
- Better workload/scheduling visibility
- Audit-ready digital records
- Data-driven enforcement and administration



LONG-TERM IMPACT



Less paperwork



Faster Verification



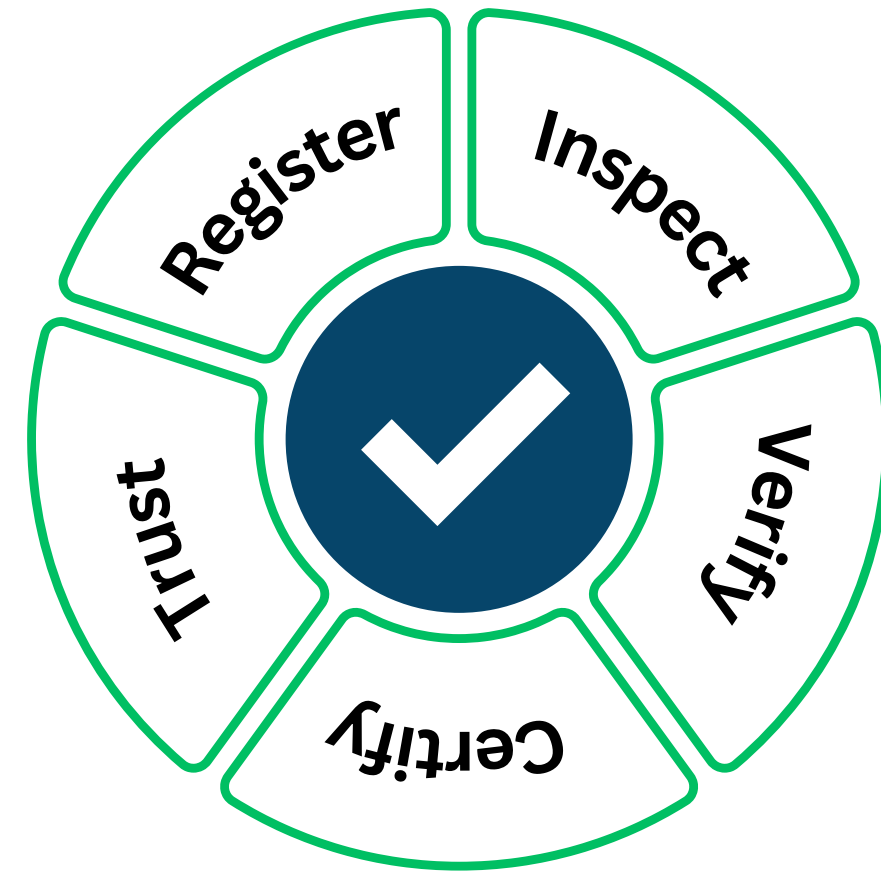
Better Traceability



Higher Transparency



Improved Consumer Trust



MapanSetu moves Legal Metrology from fragmented record-keeping to a **trusted, traceable digital lifecycle**



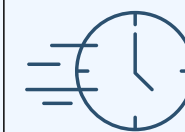
Digitalized

End-to-end digital lifecycle management



Trusted

Secure, tamper-evident and verifiable



Efficient

Reduced time, effort and manual work



Transparent

Open verification builds public confidence



- **Department of Consumer Affairs — Legal Metrology**
- Provides the primary regulatory framework for weights, measures, verification, stamping, enforcement, and implementation of Legal Metrology requirements in India.[\(.link\)](#)
- **India Code — Legal Metrology Act, 2009**
- Official statutory reference defining legal requirements, verification and stamping procedures, responsibilities of authorities, offences, and enforcement provisions governing weights and measures.[\(.link\)](#)
- **Government Services Portal — Verification & Re-verification of Weights and Measures**
- Provides the official government service workflow and requirements for verification and re-verification of weights and measuring instruments before their regulated use.[\(.link\)](#)
- **Department of Consumer Affairs — Government Approved Test Centres**
- Provides regulatory information and rules governing Government Approved Test Centres responsible for authorized testing and verification activities under the Legal Metrology framework.[\(.Link\)](#)

- **OIML R 76 — Non-Automatic Weighing Instruments**
- Provides internationally recognized technical requirements and metrological principles for non-automatic weighing instruments, supporting standardized and reliable measurement practices.[\(.link\)](#)
- **Guidelines for Indian Government**
- Aligns MapanSetu with Indian government digital-service principles, emphasizing secure platforms, accessibility, transparency, interoperability, citizen-centric services, and responsible handling of government data.[\(.link\)](#)
- **Controller of Certifying Authorities (CCA) — Digital Signatures**
- Provides the Indian regulatory framework and technical guidance for digital signatures, supporting certificate authenticity, signer verification, and integrity protection.[\(.link\)](#)
- **OWASP Application Security Verification Standard**
- Provides security verification requirements for web applications, guiding authentication, authorization, input validation, session security, data protection, and secure application development practices.[\(.link\)](#)